

low-bandwidth, high-quality voice conference calling, etc.

FreedomBox can even work as a smart on-premise router since it runs in a physical computer and can route your traffic. It can sit between various devices such as mobiles, laptops, TVs and the Internet, replacing a home wireless router. By routing traffic, FreedomBox can remove tracking advertisements and malicious Web bugs before they even reach your devices. It can cloak your location and protect your anonymity by 'onion routing' your traffic over Tor.

FreedomBox provides a VPN server that you can use while you are away from your organisation, to keep your traffic secret on untrusted public wireless networks and to securely access various devices at home. It can also be carried along with your laptop and used to connect to public networks at work or school to avail its services.

It could be used in a remote village to make available digital communications to all homes in the vicinity. In the future, FreedomBox intends to deliver support for alternative ways of connecting to the Internet such as mesh networking.

The primary design goal of FreedomBox is to be used as a personal server at home, by a single family and its friends. However, at its core, it is server software that can aid a non-technical user to set up services and maintain them with ease. Security is automatically managed and many of the technical sysadmin choices are taken care by the software, automatically, thereby reducing complexity for a non-technical user. This aspect of FreedomBox makes it well-suited for hosting services for small communities like villages or small firms.

Communities can host their own services using FreedomBox, with minimal effort. They can set up Wi-Fi networks that span the entire area of the community and

draw Internet connections from long distances. Community members can enjoy previously unavailable Internet connectivity, ubiquitous Wi-Fi coverage, free VoIP services, offline education and entertainment content, etc. This can improve privacy for individuals in the community, reduce dependence on centralised services provided by large companies, and make the community immune to censorship.

FreedomBox's features

From the apps menu, you can install applications to get the following services running from your FreedomBox.

- Access your FreedomBox from the public Internet (Pagekite, Tor or Dynamic DNS)
- Connect securely to your FreedomBox (OpenVPN)
- Chat with friends (Ejabberd, Quassel or Matrix)
- Host a conference call (Mumble)
- Publish a blog (Ikiwiki)
- Host a personal or community wiki (MediaWiki)
- Block ads while browsing the Web (Privoxy)
- Check your email (Roundcube)
- Transfer large files (Deluge, Transmission or I2P)
- Read news (Tiny Tiny RSS)
- Sync your calendar and contacts (Radicale)
- Stay connected to IRC chats (Quassel, I2P)
- Host a multi-player block sandbox (Minetest)
- Collaboratively edit a text document (infinoted)
- Keep your files synchronised to your FreedomBox (Syncthing)
- Circumvent censorship using a SOCKS5 proxy (Shadowsocks)
- Escape search engine filter bubbles using a metasearch engine (Searx)
- Privately share files (Coquelicot)

Configuring FreedomBox

From the *System* menu you can easily

configure FreedomBox with the following available options.

- Configure the system name and interface language (Configure)
- Obtain a certificate for your domain (Let's Encrypt)
- Add users and set access privileges (Users and Groups)
- Use a single login for Plinth, XMPP, Ikiwiki and SSH
- Manage network connections over Ethernet, Wi-Fi or PPPoE (Networks)
- Upgrade software packages automatically (Software Upgrades)
- Run diagnostic tests (Diagnostics)
- View which services are allowed through the firewall (Firewall)
- Configure the time zone and network time service (Date and Time)

Installing FreedomBox on Debian laptops or PCs

You can install FreedomBox in a system with Debian 9 Stretch, Debian10 Buster or newer (the latest stable version is preferred).

You should have your */etc/hosts/* set up properly before installing FreedomBox since most services offered by it require this.

If your host name is *debian-sonnet* and the FQDN (fully qualified domain name) that you are going to set to your freedombox server is *sonnet.freedombox.in*, type:

```
sudo hostnamectl set-hostname sonnet.freedombox.in
```

...and configure your */etc/hosts* properly for this, as follows:

```
127.0.0.1 localhost
127.0.1.1 sonnet.freedombox.in debian-sonnet
[you ip address] sonnet.freedombox.in
debian-sonnet
```

Now you can install FreedomBox